Last fall degree at m = 5, 6 — run report (E7, E6 completion)

Study `research/fall_degree_bounds_20261008` (aburan28/crypto). Amendment 7 (cells, gate, reading rule) was committed before any E7 draw. Amendment 8 moved the N = 54 cell to the local host before any of its draws. **Engine** `examples/hky_par.rs`, a parallel, chunked mutant closure (sha256 `24cf10d6…`). Host: Claude Code cloud container, 4 vCPU, 15 GB. Wall times are logged; the study is untimed.

Gate (on the exact E7 binary)

- It reproduces `d_F` and every per-degree closure dimension of the Sage-validated engine on 24 committed draws: m = 3 (5,2), (7,3), (11,4), (13,4), (13,5) and m = 4 (9,3). Result: `AGREE 24 DISAGREE 0`.
- It reproduces the four E6 m = 5 (10,2) draws bit for bit. They took 127–168 s, where the serial engine took 787–5264 s.

Recorded values

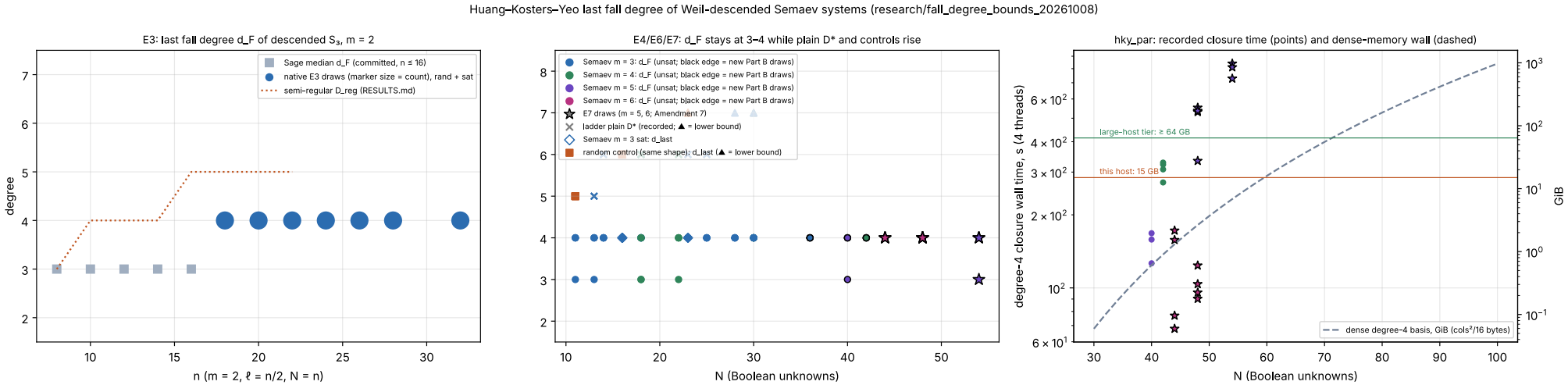


Figure 1: Left: m = 2 (E3). Middle: d_F against unknowns N for every unsatisfiable Semaev draw. Stars mark the E7 draws (m = 5 purple, m = 6 magenta). Plain-tower D* is shown as × (▲ = lower bound), and structure-matched random controls as orange squares. Right: recorded degree-4 closure wall time (points), and the dense memory that a degree-4 basis needs (dashed), against the 15 GB host and the ≥ 64 GB tier.

m	(n, ℓ)	N	columns ≤ 4	d_F	degree-4 closure, s
4	(13, 4) (E6 B completion)	42	124,314	4 4 4 4	272–328
6	(8, 2)	44	149,986	4 4 4 4	68–172
5	(11, 3)	48	213,053	4 4 4 4	334–553
6	(9, 2)	48	213,053	4 4 4 4	90–124
5	(13, 3)	54	342,541	4 3 4 4	728–839 (peak 12.6 GB)

Reading (Amendment 7): "d_F bounded through m = 6". All 16 E7 draws have d_F ≤ 4. Not run: (6, 10, 3) N = 58 and (5, 15, 3) N = 60, the ≥ 64 GB tier, because no host was available. They are not extrapolated. N ≈ 100 needs about 4.1M columns (a ≈ 1 TB dense basis), so it needs a sparse method.

Checks

- Lean `lean/FallDegree/Verdict.lean`, kernel `decide`, 206 rows. All earlier verdicts still hold, and `e7Bounded = true`, `e7Grows = false`. Changing one E7 draw to 5 fails the build. CI regenerates `Data.lean` from the records.

- cairn: result objective `sha256:16fc1da8…` was accepted by the kernel. A first attempt was "unavailable" because `lean` was not on PATH; it is logged. `cairn audit`: chain intact.

Scope

Toy sizes only ($m \le 6$, $N \le 54$). This is not a bound. HKYY's conditional requires $d_F = \omega(m^2)$ at $m \approx n^{1/3}$ unless binary ECDLP is subexponential. Conditional crossover model (`e6/crossover_model.md`): even if $d_F = 4$ held at every size, IC would beat rho only from $n \approx 233$–283. No exponent, speedup or security claim.